So — as in the other orientation — we get six indecomposable representations of A_3:

$$1 \longrightarrow 0 \longleftarrow 0\,, \quad 0 \longrightarrow 0 \longleftarrow 1\,, \quad 1 \xrightarrow{\sim} 1 \xleftarrow{\sim} 1\,,$$

$$0 \longrightarrow 1 \longleftarrow 0\,, \quad 1 \longrightarrow 1 \longleftarrow 0\,, \quad 0 \longrightarrow 1 \longleftarrow 1\,.$$

6.3. Indecomposable representations of the quiver D_4

As the last — slightly more complicated — example we consider the quiver D_4.

Example 6.3.1 (D_4)**.** We restrict ourselves to the orientation

$$\begin{tikzcd} \bullet \arrow[r] & \bullet & \bullet \arrow[l] \\ & \bullet \arrow[u] & \end{tikzcd}.$$

So a representation of this quiver looks like

$$\begin{tikzcd} \underset{V_1}{\bullet} \arrow[r, "A_1"] & \overset{V}{\bullet} & \underset{V_3}{\bullet} \arrow[l, "A_3"'] \\ & \underset{V_2}{\bullet} \arrow[u, "A_2"'] & \end{tikzcd}.$$

The first thing we can do is — as usual — split away the kernels of the maps A_1, A_2, A_3. More precisely, we split away the representations

$$\begin{tikzcd} \underset{\ker A_1}{\bullet} \arrow[r, "0"] & \overset{0}{\bullet} & \underset{0}{\bullet} \arrow[l] \\ & \underset{0}{\bullet} \arrow[u] & \end{tikzcd},$$

$$\begin{tikzcd} \underset{0}{\bullet} \arrow[r] & \overset{0}{\bullet} & \underset{0}{\bullet} \arrow[l] \\ & \underset{\ker A_2}{\bullet} \arrow[u, "0"'] & \end{tikzcd}.$$